

Chapter 03 Homework — Rings

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Problem 03.1. Suppose that S is a set and R is a ring. Let R^S denote the set of all mappings from S to R . Show that R^S is a ring, under the operations defined by

$$(f + g)(s) = f(s) + g(s), \quad (fg)(s) = f(s)g(s).$$

Show that if S has more than one element then there exist non-zero elements f and g in R^S for which $fg = 0$.

Solution. Write $f_1, f_0 \in R^S$ for the functions

$$f_1(s) = 1_R, \quad f_0(s) = 0_R,$$

and clearly, by the specified function properties, we have all ring properties.

Prove $(f + g)h = fh + gh$. Let $s \in S$. Show $((f + g)h)(s) = (fh)(s) + (gh)(s)$:

$$((f + g)h)(s) = (f(s) + g(s))h(s) = f(s)h(s) + g(s)h(s) = (fh)(s) + (gh)(s).$$

If $|S| > 1$, there exist $s_1, s_2 \in S$ where $s_1 \neq s_2$. Let

$$f(s) = \begin{cases} 1_R & s = s_1 \\ 0_R & \text{else} \end{cases} \quad g(s) = \begin{cases} 1_R & s = s_2 \\ 0_R & \text{else} \end{cases}$$

Then $f(s)g(s) = 0$, so they are zero divisors.

Problem 03.2. Suppose that R is an integral domain, with field of fractions F . Show that the field of fractions of $R[x_1, \dots, x_n]$ can be identified naturally with $F(x_1, \dots, x_n)$.

Problem 03.3. *[TODO: statement not yet transcribed]*

Problem 03.8. Suppose that K is a field. If $f = a_0 + a_1x + \dots + a_nx^n \in K[x]$ and $k \in K$, let $(\Phi(f))(k) = a_0 + a_1k + \dots + a_nk^n$. Show that Φ is a ring homomorphism from $K[x]$ to K^K . Show that if K is finite then Φ is an epimorphism, but not a monomorphism. What happens if K is infinite?

Problem 03.10. *[TODO: statement not yet transcribed]*

Problem 03.11. *[TODO: statement not yet transcribed]*

Problem 03.12. *[TODO: statement not yet transcribed]*

Problem 03.14. *[TODO: statement not yet transcribed]*

Problem 03.15. *[TODO: statement not yet transcribed]*

Problem 03.17. *[TODO: statement not yet transcribed]*

Problem 03.25. *[TODO: statement not yet transcribed]*

Problem 03.26. *[TODO: statement not yet transcribed]*